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Assignment 4

Transactional Management

① What is transaction in DBMS

Ans A transaction in a DBMS is a logical unit of work that consists of one or more database operations such as insert, update or delete, executed as a single unit. A transaction must be completely fully or not executed at all to ensure data consistency

② Explain Acid properties

Ans Acid properties ensure reliable processing of database transactions They are.

① Atomicity: Ensures that a transaction is completely fully or not executed at all. If any parts fails, the entire transaction is rolled back.

② Consistency: Ensures that the database remains in a valid state before & after the transaction. It maintains data integrity

③ Isolation: Ensures that multiple transactions are executed independently without affecting each other.

④ Durability: Ensures that once a transaction is ~~completed~~ committed the changes are permanently stored even in case of system failure.

⑤ What is concurrency control?

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Ans. Concurrency control in DBMS is the process of managing multiple transactions that are executed simultaneously without causing conflicts or inco-nsistencies in the database.
It ensures that transactions are performed in a contr-olled manner so that the integrity and correctness of data are maintained.

④ What is a deadlock?

Ans. A deadlock is a situation in a DBMS where two or more transactions are waiting indefinitely for each other to release resources, and none of them can proce
This results in a standstill where the transactions cannot complete their execution.

⑤ What is serializability?

Ans. It is a property of a database system that ensures the results of executing multiple transactions concurrently is the same as if the transactions were executed one after another (in a serial order).
It helps maintain data consistency and correctness in concurrent transaction processing.

⑥ What is recovery in DBMS?

Ans. Recovery in DBMS is the process of restoring the database to a correct and consistent state after a failure, such as system crash, hardware failure or software error.
It ensures that no data is lost and that all committed transactions are preserved while incomplete transact-ons are rolled back.

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(7) What is a checkpoint?

Ans. A checkpoint in DBMS is a mechanism that saves the current state of the database at a specific point in time. It helps reduce recovery time by allowing the system to restart from the last checkpoint instead of processing all transactions from the beginning after a failure.

Design, Analyze and Defend

(8) Explain ACID properties using a banking example.

Ans. Consider a bank transaction where ₹1000 is transferred from Account A to Account B.

- **Atomicity**:- The transaction must be completed fully - ₹1000 is deducted from Account A and added to Account B. If any step fails, the whole transaction is cancelled and no money is transferred.
- **Consistency**:- Before and after the transaction, the total balance remains correct. If A had ₹5000 and B had ₹3000, the total ₹8000 remains the same after transfer.
- **Isolation**:- If multiple transactions occurs at the same time, each transaction is processed independently without affecting others. No other transaction can see incomplete changes.
- **Durability**: once the transaction is completed successfully, the changes (money transfer) are permanently saved, even if the system crashes afterward.

(9) Illustrate a deadlock situation with two transactions.

Ans. A deadlock occurs when two transactions are waiting for each other to release resources, and neither can proceed.

Example:

- Transaction T1 locks Resource A.

- Transaction T2 locks Resource B.

Now:

- T1 tries to lock Resource B (but it is already locked by T2).
- T2 tries to lock Resource A (but it is already locked by T1)

Both transactions keep waiting for each other indefinitely, resulting in a deadlock.

Conclusion:- Neither T1 and T2 can continue execution because each is waiting for a resource held by the other.

(10) Explain serial V/s concurrent execution with example.

Ans: Serial Execution:- In serial execution, transactions are executed one after another, without any overlap. A new transaction starts only after the previous one is completed.

Concurrent Execution:- In concurrent execution, multiple transactions are executed at the same time, with their operations interleaved to improve performance and resource utilization.

Example:- Consider two transactions:-

- T1:- Transfer ₹500 from Account A to B
- T2:- withdraw ₹200 from Account A
- In serial execution, T1 completes first then T2 starts.
- In concurrent execution, operations of T1 and T2 may run simultaneously (e.g. deduction steps overlap), but the final result should be the same as some serial order.

(11) A banking system transfers money between accounts. Explain how acid properties apply.

Ans. In a banking system, when money is transferred

from one account to another, ACID properties ensures safe and reliable transactions.

- **Atomicity**: The transfer must be fully completed. money is deducted from one account and added to another. If any step fails, the entire transaction is cancelled.
- **Consistency**: The total balance remains correct before and after the transaction. The database follows all rules and constraints.
- **Isolation**: Multiple transactions happening at the same time do not interfere with each other. Each transaction behaves as if it is executed alone.
- **Durability**: once the transaction is successfully completed, the changes are permanently saved, even if the system crashes.

Conclusion: ACID properties ensure that banking transactions are accurate, secure and reliable.

Q) Two transactions update the same data simultaneously. Identify possible problems.

Ans. When two transactions update the same data at the same time several concurrency issues can occur:

- **Lost Update problem**: one transaction overwrites the changes made by another, causing loss of data.
- **Dirty Read (uncommitted data)**: A transaction reads data that has been modified by another transaction but not yet commit-ted.
- **Non-Repeatable Read**: A transaction reads the same data twice and gets different values because another transaction updated it in between.
- **Inconsistent Data (Incorrect summary)**: A transaction reads partial updates made by another transaction, leading to incorrect results.

Conclusion: These problems highlight the need for concurrency control mechanisms to maintain data consistency.

(13) A system crashes during a transaction. Explain recov-ery steps.

Ans. When a system crashes during a transaction, the DBM uses recovery mechanisms to restore the database to a consistent state.

- Recovery steps :-

(1) Check Log File :- The system examines the transaction log (log file), which records all operations performed by transactions.

(2) Identify Transactions :-

- Committed transactions (complete before crash)
- Uncommitted transactions (incomplete at the time of crash)

(3) Redo Operation (Roll Forward) :- All committed trans-actions are reapplied to ensure their changes are preserved.

(4) Undo operation (Rollback) :- All uncommitted transac-tions are reversed to remove incomplete changes.

(5) Restore to consistent state :- After redo and undo, the database is brought back to a correct and stable condition.

One concept + one Insight

(14) Explain one ACID property with a real life example.

Ans. Atomicity :- Atomicity means a transaction is complete fully or not at all.

Example :- When you withdraw money from an ATM, the system must both deduct money from your account and dispense cash. If cash is not dispensed, the amount should not be deducted. Either both actions happen or none.

(15) Draw a transaction timeline and identify conflict.

Ans. Example Timeline:-

T1	T2
R(A)	
	R(A)
W(A)	
	W(A)

Explanation:-

- T1 reads and writes data item A.
- T2 also reads and writes the same data item A at the same time.

Conflict:- A conflict occurs when two transactions access the same data and at least one operation is a write. Here, both T1 and T2 are writing to A → write-write conflict.

(16) Explain one recovery technique.

Ans. Log Based Recovery:- Log-Based recovery is a technique where all database operations are recorded in a log file before they are executed.

How it works:-

- Every transaction's actions are stored in the log.
- If a crash occurs-

→ Redo is applied for committed transactions.

→ Undo is applied for uncommitted transactions.

Result:- The database is restored to a consistent state using the log.